

SIMPLEX Central Operations Console

USER GUIDE

Client Quickstart & Installation Manual (Windows, Linux & macOS)

1-Click Docker Deployment

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Client Quickstart, Installation & Operations Manual

Official Multi-Platform Deployment Guide (Windows, Linux & macOS)

Executive Overview

This manual provides non-technical, step-by-step instructions to obtain, install, launch, update, and operate the **SIMPLEX Central Operations Console** on **Windows**, **Linux**, and **macOS**.

The application is distributed as a turnkey, self-contained containerized package using **Docker**. End-users and IT administrators do not need to manually configure Node.js, Python, or database engines. All components run in secure, isolated containers with persistent storage and strict data sandboxing.



1. How to Obtain the SIMPLEX Delivery Package (`simplex-delivery.zip`)

The delivery package `simplex-delivery.zip` is extremely lightweight (~10 KB) because it contains only the operational orchestration files, environment configs, and 1-click launchers. All heavy application binaries and database engines are pulled securely from the official registry.

Clients can obtain the package through either of the following two standard methods:

Method A: Direct Download via Official GitHub Release (Recommended)

1. Open your web browser and visit the official repository releases page: 🖱️
<https://github.com/rajeshsupport/SIMPLEX/releases>
2. Under the latest release (e.g. `v1.0.0`), click on `simplex-delivery.zip` to download it directly.
3. *Alternative command line download (Linux/Mac):*

```
curl -L -O https://github.com/rajeshsupport/SIMPLEX/raw/main/simplex-delivery.zip
```

Method B: Direct Sharing from Software Vendor

If the client is operating in a closed network, intranet, or prefers direct delivery:

- The software vendor (Rajesh / PKV Global) will share `simplex-delivery.zip` via **Email attachment**, **Google Drive / OneDrive / Dropbox secure link**, or via a **USB flash drive**.
- Save the file to your computer (e.g., `Downloads` or `Desktop`).

2. Package Contents & How Docker Pulls the Images

When you extract `simplex-delivery.zip` , you will find the following clean launcher files:

File Name	Platform	Description
<code>start.bat</code>	Windows	1-Click Launcher that verifies Docker, starts all services, and opens the browser.
<code>stop.bat</code>	Windows	1-Click Stopper to safely pause all services.
<code>update.bat</code>	Windows	1-Click Updater to pull the latest patches without losing any data.
<code>start.sh</code>	Linux & macOS	1-Command Launcher for Unix systems.
<code>stop.sh</code>	Linux & macOS	1-Command Stopper for Unix systems.
<code>update.sh</code>	Linux & macOS	1-Command Updater for Unix systems.
<code>docker-compose.yml</code>	All OS	Production service definitions for Database, API, and Web UI.
<code>init-db.sql</code>	All OS	Database security isolation script ensuring access strictly to <code>SIMPLEX_CENTRAL_DB</code> .
<code>.env</code>	All OS	Configurable ports and security tokens.
<code>CLIENT_INSTRUCTIONS.txt</code>	All OS	Quick reference text file.

How Image Download Works: When `start.bat` or `start.sh` is executed for the first time, Docker reads `docker-compose.yml` and automatically pulls the official pre-compiled Docker images from Microsoft and GitHub Container Registry. The client does not need to build, compile, or install anything manually.

3. Dedicated Database Isolation & Zero Data Exposure Guarantee

A common security requirement in enterprise and hospital environments is ensuring that the application only touches its own dedicated database and never accesses other corporate or clinical databases.

1. Single Isolated Database (`SIMPLEX_CENTRAL_DB`):

- The containerized Microsoft SQL Server instance runs in complete isolation.
- None of your vendor's development databases, test fixtures, or external records exist in this container.
- Only one database is created: `SIMPLEX_CENTRAL_DB`.

2. Strict User Sandboxing (`DENY VIEW ANY DATABASE`):

- The application connects using a restricted user `simplex_app_user` with permissions limited exclusively to `SIMPLEX_CENTRAL_DB`.
- Even if deployed on an existing shared enterprise SQL Server, the script `init-db.sql` enforces `DENY VIEW ANY DATABASE`, ensuring the application cannot see or query any other database on that server.

4. WINDOWS Installation & Operation Guide (Windows 10 / 11 / Server)

Step 4.1: Install Docker Desktop on Windows

1. Download **Docker Desktop for Windows**: 🖱️ <https://www.docker.com/products/docker-desktop/>
2. Run `Docker Desktop Installer.exe`.
3. Ensure **"Use WSL 2 instead of Hyper-V"** is checked (recommended).
4. Restart your computer if prompted.
5. Open **Docker Desktop** from the Start Menu.
6. Wait until the whale icon in the bottom-right Windows taskbar turns steady green with status: **"Engine running"**.

Step 4.2: Extract the Package

1. Right-click `simplex-delivery.zip` and select **Extract All...**
2. Choose a destination folder (e.g. `C:\SIMPLEX` or `Desktop\SIMPLEX`).
3. Click **Extract**.

Step 4.3: Launch the Application (1-Click)

1. Ensure Docker Desktop is running.
2. Inside the extracted folder, **Double-Click** `start.bat`.
3. A command window will launch, verify Docker, download the latest images automatically, and initialize the system.
4. Your default web browser will automatically open to: 🖱️ <http://localhost:5173>

Step 4.4: How to Update When a Bug-Fix or Patch is Released

When the vendor notifies you of a new update:

1. Open your `SIMPLEX` folder.
2. **Double-Click** `update.bat`.
3. Docker will pull the updated container and restart the application in 15–20 seconds.
4. **Data Guarantee:** All database records, clients, and users are stored in the persistent volume `simplex_production_db_data` and are **100% PRESERVED**.

Step 4.5: How to Stop

- Double-click `stop.bat` to safely halt the application.

5. LINUX Installation & Operation Guide (Ubuntu / Debian / RHEL)

Step 5.1: Install Docker Engine on Linux

Open your terminal and run the standard official Docker installation:

```
# 1. Update package index and install prerequisites
sudo apt-get update
sudo apt-get install -y ca-certificates curl gnupg

# 2. Add Docker official GPG key
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/doc
ker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg

# 3. Add repository to Apt sources
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.do
cker.com/linux/ubuntu \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

# 4. Install Docker Engine and Docker Compose Plugin
sudo apt-get update
sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plu
gin

# 5. Enable non-root user access (Optional but recommended)
sudo usermod -aG docker $USER
```

(Log out and log back in for user group permissions to take effect).

Step 5.2: Extract and Set Permissions

```
# Unzip delivery archive
unzip simplex-delivery.zip -d ~/simplex
cd ~/simplex

# Grant executable permissions to shell scripts
chmod +x *.sh
```

Step 5.3: Launch the Application (1-Command)

```
./start.sh
```

Open your web browser and navigate to: 🖱️ <http://localhost:5173> (or <http://<server-ip>:5173> if accessing over LAN/Server).

Step 5.4: How to Apply Updates / Bug Fixes

```
cd ~/simplex
./update.sh
```

Pulls updated containers and restarts seamlessly. All database data remains 100% intact.

Step 5.5: How to Stop

```
cd ~/simplex
./stop.sh
```

6. macOS Installation & Operation Guide (Apple Silicon & Intel)

Step 6.1: Install Docker Desktop for Mac

1. Download **Docker Desktop for Mac**: 🖱️ <https://www.docker.com/products/docker-desktop/>
 - Select "**Mac with Apple Silicon**" for M1 / M2 / M3 / M4 Macs.
 - Select "**Mac with Intel chip**" for Intel-based Macs.
2. Open the downloaded `.dmg` file and drag **Docker** to your **Applications** folder.
3. Open **Docker** from Applications and accept the agreement.
4. Wait until the whale icon in the top menu bar indicates "**Docker Desktop is running**".

Step 6.2: Extract the Package

1. Double-click `simplex-delivery.zip` in Finder to extract it.
2. Open **Terminal** (`Cmd + Space` , type `Terminal` , press Enter).
3. Navigate to the extracted folder:

```
cd ~/Downloads/simplex-delivery # (or your extracted folder path)
chmod +x *.sh
```

Step 6.3: Launch the Application

```
./start.sh
```

Open your web browser to: 🖱️ <http://localhost:5173>

Step 6.4: How to Apply Updates / Bug Fixes

```
./update.sh
```

Step 6.5: How to Stop

```
./stop.sh
```

7. Default Login Credentials & Access Points

Service / Component	URL / Endpoint	Default Credentials	Description
Web Operations Portal	<code>http://localhost:5173</code>	Username: <code>admin</code> Password: <code>Rajesh@123</code>	Main dashboard for operations, users, and clients.
Backend REST API	<code>http://localhost:3000/api/v1</code>	Bearer Token / Session	Orchestration and management backend.
Swagger API Documentation	<code>http://localhost:3000/api/docs</code>	Public / Authorize Header	Interactive API specification and test console.
Database Engine	<code>localhost:1433</code>	Host: <code>simplex_db</code> Database: <code>SIMPLEX_CENTRAL_DB</code>	Dedicated Microsoft SQL Server 2022 instance.

[!NOTE] Upon your first login, it is strongly recommended to navigate to **Settings -> Security** and update the default administrator password.

8. 1-Click Over-The-Air Update Mechanism (Zero Data Loss)

When software enhancements or bug fixes are deployed:

1. The developer pushes verified code changes to GitHub.
2. GitHub Actions automatically packages and pushes the new container image to GitHub Container Registry.
3. The client executes `update.bat` (Windows) or `./update.sh` (Linux/Mac).
4. Docker detects the new image tag, downloads only the updated binary layers, and restarts the containers.
5. **Persistent Data Assurance:** The Microsoft SQL database volume (`simplex_production_db_data`) is decoupled from the application containers. User accounts, clients, audit logs, and configurations are **never overwritten or deleted** during updates.

9. Frequently Asked Questions (FAQ) & Troubleshooting

Q1: When running `start.bat`, it reports "Docker is not running". What should I do?

- **Solution:** Docker Desktop has not completed its startup. Open Docker Desktop and verify the status indicator in the bottom-left corner is green ("Engine running"). Then re-run `start.bat`.

Q2: Can multiple operators access the portal from different computers on the same network?

- **Solution:** Yes! Find the LAN IP address of the host machine (e.g. `192.168.1.100` via `ipconfig` on Windows or `ifconfig` on Linux/Mac). Other operators on the same network can access the portal via `http://192.168.1.100:5173`.

Q3: What if port 5173 or 3000 is already in use by another application?

- **Solution:** Open the `.env` file in any text editor.
 - Change `WEB_PORT=5173` to `WEB_PORT=8080` (or any available port).
 - Change `API_PORT=3000` to `API_PORT=3001`.
 - Save the file and execute `start.bat` (or `./start.sh`). The portal will be available on the new port.

Q4: Can I run this in an offline (air-gapped) environment without internet?

- **Solution:** Yes. The vendor can export the pre-built Docker containers into a single `.tar.gz` bundle (`docker save`). You can import it on the offline machine using `docker load -i simplex-offline-images.tar.gz` and then run `start.bat` without internet connectivity.

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